

Kuang-Da Wang

[LinkedIn](#)[GitHub](#)[Personal Page](#)[Google Scholar](#)

Education

 **National Yang Ming Chiao Tung University**, [ADSL](#) and [RL & Bandits Lab \(REAL\)](#) Hsinchu, Taiwan

Ph.D. candidate in Computer Science and Engineering (Advisor: Wen-Chih Peng and Ping-Chun Hsieh) Sep 2023 - Sep 2027 (Expected)

- Research topics: Sport Data Science, Imitation Learning, Reinforcement Learning and LLM Alignment
- CGPA: 3.83/4.00, 10+ publications

 **National Yang Ming Chiao Tung University**, [Advanced Database System Laboratory \(ADSL\)](#) Hsinchu, Taiwan

M.S. in Networking Engineering (Advisor: Wen-Chih Peng) Sep 2021 - Aug 2023

- CGPA: 3.90/4.00, 1 Academic Excellence Award, 2 publications

 **National Cheng Kung University** Tainan, Taiwan

B.S. in Engineering Science Sep 2018 - Jun 2021

- CGPA: 3.89/4.00, 3 Academic Excellence Award, 1 project competition award

Experience

Visiting Researcher, Carnegie Mellon University (CMU), [WAV Lab](#), May 2026 - May 2027 (Expected)

Speech Translation / Automatic Speech Recognition / Test-Time Optimization

- Developed a reward-guided latent-space test-time optimization framework for multilingual ASR that improves frozen speech foundation models through fine-grained error localization and reference-free latent search.

LLM Research Intern, Sony Group Corporation (SGC), Aug 2025 - Feb 2026

Multi-Modal LLM Agent / Multi-Agent System for Digital Design

- Designed a scalable pipeline for structured webpage data collection, multimodal generation, and section-level evaluation, enabling fine-grained assessment of LLM-generated web designs.
- Proposed an agentic model-predictive control framework for multi-turn visual-design clarification that balances design alignment with user cognitive cost.
- Developed an agentic semantic exploration framework for automatically discovering visual prompts that mitigate perception failures in large vision-language models.

PhD Researcher, National Yang Ming Chiao Tung University (NYCU), ADSL and REAL, Sep 2023 - Now

Reinforcement Learning/Imitation Learning/LLM Alignment

- Research on reinforcement learning, imitation learning, LLMs alignment, and sport strategies simulation.
- Collaborate with undergraduate and master's students on applied machine learning research.
- Establish robust evaluation metrics for book-level (long-form) translation, focusing on global consistency and contextual alignment.

AI Intern, Taiwan Semiconductor Manufacturing Company (TSMC), AI4BI, Jun 2023 - Dec 2024

Document Intelligence/Multi-Agent LLM Framework/Information Retrieval

- A comparative analysis of agentic and decomposed information retrieval methods for financial report generation.
- Develop applications for tabular information extraction from image-based financial documents.

Projects

Test-Time Alignment and Reasoning for Large Language Models, Jun 2024 - Present

- Develop a novel test-time alignment framework that treats LLM alignment as sequential decision-making, balancing the curse of horizon (token-level) and curse of dimensionality (response-level).

- Converting intermediate reasoning into sequences of sub-questions and sub-answers, and using QA-level self-verification as dense rewards over reasoning trajectories to enhance reasoning performance at test-time.

Badminton Players' Strategies Simulation and Enhancement, [Precision Sport Science](#) Feb 2022 - Present

- Develop a realistic badminton environment with lifelike opponents and physics simulation for quick strategy validation and advanced tactical assessment.
- Build LLM-based badminton agents and explore language-based strategy simulations.

Selected Publications

Masao Someki, Samuele Cornell, **Kuang-Da Wang**, Shinji Watanabe, "ATLAS: Agentic ASR Tuning for Research Labor Scaling", **SLT 2027** [[TBD](#)]

Kuang-Da Wang*, Kai-Jie Lin*, Guo-Xun Ko, Yu-Heng Lin, Yu Heng Hung, Travis M. Bartley, Shuoyang Ding, Yu-Chiang Frank Wang, Guan-Ting Liu, Chao-Han Huck Yang, Wen-Chih Peng, Ping-Chun Hsieh, "RLVR from Within: Learning to Ask to Elicit Intrinsic Rewards for Test-Time Reasoning", **ICLR 2027** (under review) [[TBD](#)]

Kuang-Da Wang*, Kai-Jie Lin*, Guo-Xun Ko, Yu-Heng Lin, Yu Heng Hung, Travis M. Bartley, Shuoyang Ding, Yu-Chiang Frank Wang, Chao-Han Huck Yang, Wen-Chih Peng, Ping-Chun Hsieh, "AskR: Learning to Ask for Intrinsic Reward Elicitation for Test-Time Reasoning", **ICLR 2027** (under review) [[TBD](#)]

Jaechang Kim, Yotaro Shimose, Zhao Wang, **Kuang-Da Wang**, Jungseul Ok, Shingo Takamatsu, "Visual Prompt Discovery via Semantic Exploration", **ECCV 2026** [[paper](#)] (Work done as an intern at Sony Group Corporation)

Kuang-Da Wang*, Zhao Wang*, Wei-Yao Wang, Yotaro Shimose, Jaechang Kim, Shingo Takamatsu, "Agentic Model Predictive Questioning Control in Visual Design", **ICML 2026** [[TBD](#)] (Work done as an intern at Sony Group Corporation)

Kuang-Da Wang*, Zhao Wang*, Yotaro Shimose, Wei-Yao Wang, Shingo Takamatsu, "WebGen-V Bench: Structured Representation for Enhancing Visual Design in LLM-based Web Generation and Evaluation", **KDD 2026** [[paper](#)] (Work done as an intern at Sony Group Corporation)

Yen-Che Chien*, **Kuang-Da Wang***, Wei-Yao Wang, Wen-Chih Peng, "Benchmarking Agentic Newswriting via Journalistic Workflows", **ACL Findings 2026** [[paper](#)]

Yung Chien Wang, **Kuang-Da Wang**, Wei-Yao Wang, Wen-Chih Peng, "Adapting to Evolving Data: Test-Time Expert Aggregation for Imbalanced Tabular Regression", **WSDM 2026 Short Paper** [[paper](#)]

Kuang-Da Wang, Teng-Ruei Chen, Yu Heng Hung, Guo-Xun Ko, Shuoyang Ding, Yueh-Hua Wu, Yu-Chiang Frank Wang, Chao-Han Huck Yang, Wen-Chih Peng, Ping-Chun Hsieh, "Test-Time Alignment for Large Language Models via Textual Model Predictive Control", **ICLR 2026** [[paper](#)] (This work was conducted in collaboration with NVIDIA)

Kuang-Da Wang, Shuoyang Ding, Chao-Han Huck Yang, Ping-Chun Hsieh, Wen-Chih Peng, Vitaly Lavrukhin, Boris Ginsburg "Extending Automatic Machine Translation Evaluation to Book-Length Documents", **EMNLP main 2025** [[paper](#), [GitHub](#)] (This work was conducted in collaboration with NVIDIA)

Yang Chang, **Kuang-Da Wang**, Ping-Chun Hsieh, Cheng-Kuan Lin, Wen-Chih Peng, "DDOT: A Derivative-directed Dual-decoder Ordinary Differential Equation Transformer for Dynamic System Modeling", **PAKDD 2025** [[paper](#)].

Yong-En Tian, Yu-Chien Tang, **Kuang-Da Wang**, An-Zi Yen and Wen-Chih Peng, "Template-Based Financial Report Generation in Agentic and Decomposed Information Retrieval", **SIGIR 2025 Short Paper** [[paper](#), [GitHub](#)]

Peng, Wen-Chih; **Wang, Kuang-Da**; Xie, Bo-Zhou; Chen, Yu-An; Tsai, Cheng-Shiuan; Doong, Shao-Jyun; Hung, Jun-Chen, "CoachAI+ Badminton Environment: Realistic Badminton Game Simulator for Enhancing Player Performance", **The MIT Sloan Sports Analytics Conference 2025 Research Papers Competition Finalist** [[paper](#), [GitHub](#), [presentation video](#)].

Bing-Zhi Ke, **Kuang-Da Wang**, Wen-Chih Peng, "A Representation-Guided Diffusion Model Framework for Strategic Planning in Badminton", **AAMAS 2025 Extended Abstract** [[paper](#)].

Kuang-Da Wang, Wei-Yao Wang, Ping-Chun Hsieh, Wen-Chih Peng, "Offline Imitation of Badminton Player Behavior via Experiential Contexts and Brownian Motion", **ECML-PKDD 2024** [[paper](#)].

Kuang-Da Wang, Yu-Tse Chen, Yu-Heng Lin, Wei-Yao Wang, Wen-Chih Peng, "The CoachAI Badminton Environment: Bridging the Gap between a Reinforcement Learning Environment and Real-World Badminton Games", **AAAI 2024** [[paper](#)].

Awards

[Nov. 2023] Master Thesis Award, Taiwanese Association for Artificial Intelligence

[Sep. 2023] Outstanding Doctoral Student Scholarship, Ministry of Science and Technology